

PRODUCT REQUIREMENTS DOCUMENT

Smart Attendance System

QR Code & Face Recognition Attendance Platform

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1. Introduction

1.1 Purpose

This Product Requirements Document (PRD) defines the complete specifications, features, user flows, and system behaviours for the Smart Attendance System — a web-based platform that enables educational institutions to record and manage student attendance using two methods: QR Code scanning and Face Recognition. The document is intended to serve as a complete blueprint for design, development, testing, and deployment of the system.

1.2 Project Overview

The Smart Attendance System replaces manual roll-calls and paper-based registers with a secure, fast, and intelligent digital solution. Lecturers can start an attendance session, generate a time-limited QR code and a session verification code, and students can mark themselves present by scanning their face or scanning the QR code — after entering the session code to confirm physical presence in the classroom.

1.3 Goals & Objectives

- Eliminate proxy attendance and fraudulent check-ins.
- Provide real-time attendance tracking for lecturers and administrators.
- Automate report generation, threshold alerts, and notifications.
- Support two attendance methods (QR Code and Face Recognition) with seamless fallback.
- Make the system accessible entirely from a web browser — no native app required.

1.4 System Roles

Role	Description	Access Level
Admin	Institution-level manager. Sets up the entire system structure.	Full Access
Lecturer	Course facilitator. Creates sessions and monitors attendance.	Course-level Access
Student	End user. Marks own attendance via QR or Face scan.	Personal Access Only

2. Admin Module

The Admin is the super-user of the system. The admin sets up the institutional structure (departments, programmes, courses), manages all user accounts, and has visibility into every report and activity across the platform. There is only one admin account per institution, created during the initial system setup.

2.1 Admin Account Creation & First Login

2.1.1 System Initialisation

When the system is deployed for the first time, a one-time setup screen is displayed. The admin provides the following to create the institution account:

- Institution name
- Institution logo (optional upload)
- Admin full name
- Admin official email address
- Admin password (must meet strength requirements)
- Academic year format (e.g. 2024/2025)

After setup, the admin is taken to their dashboard and the setup screen is permanently disabled.

2.1.2 Admin Login

The admin logs in at the main login page using their registered email and password. On successful authentication, the system issues a JWT token and redirects the admin to the Admin Dashboard. Failed login attempts are counted, and after 5 consecutive failures the account is temporarily locked for 15 minutes.

2.2 Admin Dashboard

The Admin Dashboard is the central control panel. It presents a real-time overview of the institution and quick access to all management modules.

2.2.1 Dashboard Overview Cards

- Total number of registered students
- Total number of lecturers
- Total number of active courses this semester
- Total number of departments
- Total attendance sessions held today
- Number of students below the attendance threshold institution-wide

2.2.2 Quick Action Shortcuts

- Add New Lecturer
- Add New Student
- Create New Course
- View All Reports

2.2.3 Recent Activity Feed

A live log showing recent system events, such as new students registered, sessions started, and reports exported.

2.3 Department Management

Departments are the top-level organisational unit. All programmes and courses belong to a department.

2.3.1 Creating a Department

The admin navigates to Settings > Departments > Add Department and fills in:

- Department name (e.g. Department of Computer Science)
- Department code (e.g. DCS)
- Faculty or school it belongs to (e.g. Faculty of Engineering)

The department is saved and immediately available for programme assignment.

2.3.2 Managing Departments

- Edit department name, code, or faculty at any time.
- View all programmes under a department.
- Deactivate a department (hides it from new entries but preserves historical data).
- Delete a department only if it has no programmes assigned.

2.4 Programme Management

A programme is an academic course of study (e.g. BSc Computer Science). Programmes belong to a department and are broken into levels/years.

2.4.1 Creating a Programme

Admin navigates to Academics > Programmes > Add Programme and fills in:

- Programme name (e.g. BSc Computer Science)
- Programme code (e.g. BSCS)
- Department it belongs to
- Duration in years (e.g. 4 years)
- Levels/Years (auto-created based on duration — e.g. Level 100, 200, 300, 400)

2.4.2 Managing Programmes

- Edit programme details at any time.
- View all students enrolled in a programme per level.

- View all courses assigned to a programme per level.
- Deactivate or archive a programme.

2.5 Course Management

Courses are the individual units of study within a programme. A course belongs to a programme, is assigned to a level, and is taught by a lecturer.

2.5.1 Creating a Course

Admin navigates to Academics > Courses > Add Course and fills in:

- Course title (e.g. Database Systems)
- Course code (e.g. CS301)
- Programme it belongs to
- Level/Year it is taught in (e.g. Level 300)
- Semester (First or Second)
- Credit hours
- Assigned lecturer (selected from the lecturer list)
- Minimum attendance threshold percentage (default: 75%)

2.5.2 Course-Programme Auto-Assignment

When a course is created and assigned to a programme and level, every student currently enrolled in that programme and level is automatically linked to the course. New students enrolled in that programme and level in the future are also automatically linked.

2.5.3 Managing Courses

- Edit any course detail at any time.
- Reassign a course to a different lecturer.
- View all students enrolled in a course.
- View full attendance history for a course.
- Clone a course to a new semester.
- Deactivate a course at the end of a semester.

2.6 Lecturer Management

The admin is the only one who can create lecturer accounts. Lecturers cannot self-register.

2.6.1 Adding a New Lecturer

Admin navigates to Users > Lecturers > Add Lecturer and enters:

- Full name
- Official staff email address
- Staff ID number
- Department
- Phone number (optional)

The system immediately sends an email to the lecturer with a secure activation link (valid for 72 hours). The lecturer clicks the link and sets their own password to activate their account. The admin does not set or know the lecturer's password.

2.6.2 Assigning Courses to a Lecturer

Courses are assigned to a lecturer during course creation (Section 2.5.1), or at any time by editing the course. A lecturer can be assigned to multiple courses simultaneously.

2.6.3 Managing Lecturer Accounts

- View all courses assigned to a lecturer.
- View attendance sessions a lecturer has conducted.
- Reassign courses to a different lecturer.
- Temporarily suspend a lecturer account (blocks login).
- Permanently delete a lecturer account (only if no active courses assigned).
- Reset a lecturer's password (triggers a password reset email to the lecturer).

2.7 Student Management

The admin registers students in bulk or individually. Students complete their own onboarding after the admin initiates their account.

2.7.1 Adding a Student (Individual)

Admin navigates to Users > Students > Add Student and enters:

- Full name
- Student ID number
- Official student email address
- Department
- Programme
- Level/Year of study
- Semester of entry

After saving, the system automatically:

1. Links the student to all courses under their programme and level.
2. Sends an invitation email with a secure registration link (valid for 48 hours).

2.7.2 Bulk Student Import

For large cohorts, the admin can upload a CSV or Excel file containing student records. The system validates each row, creates accounts, and sends invitation emails in bulk. A summary report shows how many accounts were created successfully and which rows had errors.

2.7.3 Student Onboarding (Completed by the Student)

When the student receives the invitation email and clicks the link, they are taken to a registration completion page where they:

3. Set a secure password for their account.
4. Confirm their email address.

5. Upload a clear frontal face photo for the face recognition system.

The uploaded photo is processed by the backend. The face is detected, a unique face encoding (numerical vector) is extracted and stored in the database. The raw photo is discarded after encoding. If no face is detected in the photo, the student is prompted to re-upload.

2.7.4 Managing Student Accounts

- View a student's full profile including their courses and attendance history.
- Edit student details (name, email, level, programme).
- Move a student to a new level at the start of a new academic year.
- Suspend or deactivate a student account.
- Manually mark or adjust attendance records (with audit log).
- Re-send invitation email if the student did not receive or link expired.
- Reset a student's password.

2.8 Academic Year & Semester Management

- Create and manage academic years (e.g. 2024/2025).
- Set the current active academic year.
- Define semesters within each academic year (e.g. Semester 1, Semester 2).
- Activate or close a semester — closing a semester archives all attendance data for that period.

2.9 Attendance Threshold Settings

The admin sets the institution-wide default minimum attendance percentage (default: 75%). This can be overridden per course during course creation. When a student's attendance in a course falls below the threshold, the system automatically sends them an email warning. The admin can configure how often warnings are sent (e.g. at 80%, 75%, and 70%).

2.10 Reports & Analytics (Admin)

2.10.1 Available Reports

Report	Description	Export Format
Institution-wide Attendance	Overall attendance across all courses and departments.	PDF, Excel
Per-Department Report	Attendance breakdown for all courses in a department.	PDF, Excel
Per-Course Report	Full attendance record for a single course.	PDF, Excel
Per-Student Report	Individual student attendance across all their courses.	PDF

Report	Description	Export Format
Defaulters Report	All students below the attendance threshold.	PDF, Excel
Lecturer Activity Report	Sessions created and attendance rates per lecturer.	PDF, Excel

2.10.2 Dashboard Charts

- Bar chart: attendance rate per department.
- Line chart: attendance trends over the semester week by week.
- Pie chart: present vs absent ratio institution-wide.
- Table: top 10 courses with lowest attendance rates.

2.11 Notifications & Alerts (Admin)

- Email notification when a student falls below the attendance threshold.
- Weekly summary email to the admin with institution attendance overview.
- Alert when a lecturer has not conducted any session in over 2 weeks.
- Alert when a student account invitation has expired without being used.

2.12 System Settings

- Update institution name and logo.
- Change admin email or password.
- Configure SMTP email settings for system notifications.
- Set the session code length (default: 6 characters) — see Section 4.3.
- Set the default QR code expiry duration (default: 15 minutes).
- Set the face recognition confidence threshold (default: 80%).
- View system logs and audit trail of all admin actions.
- Manage academic year and semester cycles.

3. Lecturer Module

The Lecturer is responsible for managing their courses and conducting attendance sessions. Lecturers cannot create their own accounts or courses — these are set up by the Admin. A lecturer interacts with the system primarily through their course dashboard and session management tools.

3.1 Lecturer Account Activation & Login

3.1.1 Account Activation

A lecturer receives an email from the system with a unique activation link after the admin adds them. The lecturer clicks the link and is directed to an activation page where they set their password. The activation link expires after 72 hours. If expired, the lecturer must contact the admin to resend the link.

3.1.2 Lecturer Login

6. The lecturer goes to the system login page.
7. Enters their staff email and password.
8. Clicks Sign In.
9. On success, a JWT token is issued and the lecturer is redirected to their dashboard.
10. On failure (wrong credentials), an error message is shown. After 5 failed attempts, the account is locked for 15 minutes.
11. A Forgot Password link sends a password reset email.

3.2 Lecturer Dashboard

The Lecturer Dashboard provides a quick view of all assigned courses and recent activity.

- List of all courses assigned to the lecturer with the number of enrolled students.
- Upcoming or recently active sessions.
- Overall attendance rate per course (shown as percentage badges).
- Number of students below the attendance threshold per course.
- Quick button to start a new attendance session for any course.

3.3 Course View

The lecturer clicks on any course from their dashboard to open the Course View. This screen shows:

- Course title, code, programme, level, and semester.
- Full list of enrolled students with their photo, name, ID, and current attendance percentage.
- Visual indicator (green/amber/red) per student based on their attendance standing.

- History of all past attendance sessions for this course.
- Button to start a new attendance session.
- Button to export the attendance report for this course.

3.4 Creating & Running an Attendance Session

This is the core function of the lecturer module. A session represents a single class meeting for which attendance is being recorded.

3.4.1 Starting a Session

12. The lecturer opens a course and clicks Start Attendance Session.
13. A session configuration panel appears with the following fields:
 - Session label/title (optional, e.g. Week 5 Lecture)
 - Session date and time (auto-filled with current date/time, editable)
 - QR code expiry duration (default set by admin, lecturer can override — e.g. 10, 15, or 20 minutes)
14. The lecturer clicks Create Session.
15. The system generates two security items:
 - A unique, time-limited QR Code tied to the session.
 - A 6-character alphanumeric Session Verification Code (e.g. AX72KC).
16. The session is now live and active.

3.4.2 The Active Session Screen

The active session screen shows:

- The QR Code displayed prominently for students to scan (the lecturer projects this on a screen).
- The Session Verification Code displayed clearly below the QR code (the lecturer announces this verbally or writes it on the board — only those physically present will know it).
- A live countdown timer showing how long until the QR code expires.
- A live list of students who have successfully marked attendance — updating in real time as students check in.
- The total count of present students vs total enrolled students.
- A Refresh QR Code button (generates a new QR code without ending the session, useful if the QR expires before all students check in).
- An End Session button.

3.4.3 Ending a Session

17. The lecturer clicks End Session.
18. A confirmation prompt appears: Are you sure? This will finalise attendance.
19. On confirmation, the system:
 - Invalidates the QR code and the session verification code.

- Marks all students who did not check in as Absent.
 - Locks the session so no further check-ins are possible.
20. The session summary page is displayed, showing total present, total absent, and the check-in method used by each student.

3.4.4 Post-Session Management

- The lecturer can view the full attendance record of any past session.
- The lecturer can manually override a student's attendance status for a session (e.g. mark Present if a student had a valid technical issue), with a mandatory reason note. All manual overrides are logged in the audit trail.
- The lecturer can delete a session entirely (only before any attendance is recorded).

3.5 Attendance Records & Reports (Lecturer)

- View attendance records for all sessions in a course, filterable by date or student.
- See each student's overall attendance percentage for the course.
- View a list of students currently below the attendance threshold.
- Export full attendance report as PDF or Excel for submission or record-keeping.
- See a bar chart of attendance rate per session over time.

3.6 Notifications (Lecturer)

- Email reminder if a session was started but not ended after a configurable timeout.
- Weekly summary email: attendance rate per course, number of students at risk.
- In-app notification when a student's attendance drops below threshold.

3.7 Profile & Account Settings (Lecturer)

- Update display name and phone number.
- Change password (requires current password confirmation).
- Update profile photo.
- Manage email notification preferences (opt in or out of weekly summary).

4. Student Module

The Student is the end user who marks their own attendance. Students cannot create their own accounts from scratch — they are initiated by the Admin and complete their registration via an email invitation.

4.1 Student Registration & Onboarding

4.1.1 Invitation Email

After the Admin adds a student to the system (Section 2.7.1), the student receives an email that contains:

- A welcome message explaining the system.
- Their pre-assigned Student ID and the courses they have been enrolled in.
- A unique, secure registration link (valid for 48 hours).
- Instructions on how to complete registration.

4.1.2 Completing Registration

The student clicks the link and is taken to the registration completion page:

21. The student's name and student ID are pre-filled and read-only (set by admin).
22. The student creates a password (minimum 8 characters, must include a number and a special character).
23. The student confirms the password.
24. The student uploads a face photo. Requirements for the photo:
 - Clear frontal view of the face with no obstructions.
 - Good lighting — not too dark or washed out.
 - Only one face visible in the frame.
 - Minimum resolution of 300x300 pixels.
25. The student clicks Complete Registration.
26. The system processes the photo — detects the face and extracts the face encoding. If no face is detected, an error is shown and the student must re-upload.
27. On success, the student is redirected to their dashboard and a confirmation email is sent.

4.1.3 Expired Invitation Link

If the student clicks the link after 48 hours, they see a message informing them the link has expired and are prompted to contact their institution admin for a new invitation link.

4.2 Student Login

28. Student goes to the system login page.
29. Enters their student email and password.
30. Clicks Sign In.
31. On success, the student is directed to their personal dashboard.
32. Forgot Password link sends a password reset email to the student.

4.3 Student Dashboard

The Student Dashboard gives the student a clear picture of their attendance status across all enrolled courses.

- List of all enrolled courses with current attendance percentage for each.
- Colour-coded attendance badge: Green (above threshold), Amber (approaching threshold), Red (below threshold).
- A Mark Attendance button — active only when there is a live session for one of their courses.
- Upcoming class schedule (if configured by admin).
- Recent attendance activity log (last 5 check-ins).

4.4 Marking Attendance — Full Step-by-Step Flow

This is the most critical part of the student experience. Below is the complete flow from start to finish.

Prerequisites

A lecturer must have started an active session for the student's course. The QR code and session verification code must still be valid.

4.4.1 Step 1 — Opening the Attendance Page

When a live session is active for one of the student's courses, a prominent notification banner appears on their dashboard: Your lecturer has started an attendance session for [Course Name]. The student clicks Mark Attendance Now or taps the notification banner.

4.4.2 Step 2 — Entering the Session Verification Code

Before the student can access the QR scanner or face scan camera, they are presented with a code entry screen:

Enter Session Code

Your lecturer has announced a 6-character code. Enter it below to confirm you are physically present in the classroom.

[_ _ _ _ _ _]
[Submit Code]

The student types the 6-character code they received from the lecturer (announced verbally or written on the board). This step ensures that only students who are physically in the classroom know the code. If the code is wrong, an error message is shown: Incorrect code. Please check with your lecturer. After 3 wrong attempts, the student is locked out of the session.

4.4.3 Step 3 — Choosing Attendance Method

After entering the correct code, the student is presented with two attendance options:

Method	Description	Best For
Face Scan	Uses the device's front camera to verify identity via face recognition.	Quick, hands-free check-in
QR Code Scan	Uses the device's back camera to scan the QR code on the lecturer's screen.	Reliable fallback option

4.4.4 Step 4A — Face Scan Method

33. The student selects Scan My Face.
34. The browser requests camera permission. The student must allow access.
35. The front-facing camera activates and a real-time video preview is shown.
36. A face detection overlay appears on the preview. The student positions their face inside the guide frame.
37. When a face is clearly detected and stable, the system automatically captures a snapshot.
38. The snapshot is sent securely to the backend server.
39. The backend compares the snapshot against the student's stored face encoding using DeepFace.
40. Possible outcomes:
 - Match confirmed (confidence $\geq 80\%$): Attendance is marked as Present (Face Scan). A success screen is shown: You are marked Present for [Course Name].
 - Match failed (confidence $< 80\%$) or camera error: The student sees a prompt: Face scan was not successful. Would you like to switch to QR Code scanning?
41. The student can tap Try Again or Switch to QR Code.

4.4.5 Step 4B — QR Code Scan Method

42. The student selects Scan QR Code (either by choice or after a failed face scan).
43. The browser requests camera permission. The student must allow access.
44. The back-facing camera activates and a scanning frame is shown.
45. The student points the camera at the QR code displayed on the lecturer's screen.

46. The QR code is detected and decoded automatically — no button press needed.
47. The decoded data is sent to the backend.
48. The backend validates the QR code:
 - Is the QR code still within its validity period?
 - Does it belong to the active session for the student's course?
 - Has this student already marked attendance for this session?
49. Possible outcomes:
 - All checks pass: Attendance is marked as Present (QR Code). A success screen is shown.
 - QR code expired: Student sees a message: This QR code has expired. Ask your lecturer to refresh it.
 - Already marked: Student sees: You have already marked attendance for this session.
 - Not enrolled: Student sees: You are not enrolled in the course for this session.

4.4.6 Step 5 — Attendance Confirmation

After a successful check-in (by either method), the student sees a confirmation screen showing:

- A green checkmark icon.
- Course name and session details.
- Date and time of check-in.
- Method used (Face Scan or QR Code).
- The student's updated attendance percentage for the course.
- A button to return to the dashboard.

4.5 Anti-Cheat & Security Mechanisms

Threat	Prevention Mechanism
Scanning QR code remotely (screenshot sharing)	QR expires in 10–15 mins. Session code must be entered first — only those in class hear the code.
Marking attendance for another student	Student must be logged in with their own account. Face scan verifies physical identity.
Submitting the QR code multiple times	Backend checks for duplicate entries per student per session.
Guessing the session verification code	Codes are randomly generated. 3 wrong attempts locks the student out.
Face spoofing (showing a photo of the student)	Liveness detection (if enabled) flags static images. Confidence threshold reduces false positives.
Marking attendance after the session ends	Session is locked when the lecturer ends it. No further check-ins accepted.

4.6 Viewing Attendance History (Student)

- The student can view a detailed attendance record for each enrolled course.
- Records show: session date, session label, check-in time, method used (Face or QR), and status (Present / Absent).
- Overall attendance percentage is shown per course with a colour-coded status.
- A warning banner is shown if the student is approaching or has fallen below the threshold.

4.7 Student Profile & Settings

- View and update display name.
- Change password (requires current password confirmation).
- Update face photo for face recognition — the new photo goes through the same validation and encoding process.
- View enrolled courses.
- Manage email notification preferences.

5. Notification System

Trigger	Recipient	Channel	Content
Student added by admin	Student	Email	Invitation with registration link
Lecturer added by admin	Lecturer	Email	Activation link and instructions
Student completes registration	Admin	Email	Confirmation notification
Attendance session started	Students (enrolled)	In-app	Banner: Session is live for [Course]
Student below 80% threshold	Student	Email	Warning: Attendance falling
Student below 75% threshold	Student + Admin	Email	Critical warning: Below minimum
Session not ended after 2 hrs	Lecturer	Email	Reminder to close session
Weekly attendance summary	Admin + Lecturers	Email	Attendance rate overview
Password reset requested	User	Email	Secure reset link (30 min expiry)

6. Technical Architecture & Stack

6.1 Technology Stack

Layer	Technology	Purpose
Frontend	React.js	UI and component rendering
Frontend Styling	Tailwind CSS	Responsive and consistent styling
HTTP Client	Axios	API communication from frontend
Face Detection (Client)	face-api.js	In-browser real-time face detection
QR Generator	qrcode.react	Generate and display QR codes
QR Scanner	html5-qrcode	Scan QR codes via browser camera
Backend	Python FastAPI	REST API server
Face Recognition (Server)	DeepFace	Match face snapshots to encodings
Image Processing	OpenCV	Pre-process captured images
Authentication	JWT (JSON Web Tokens)	Secure session management
Database	PostgreSQL	Primary relational data store
ORM	SQLAlchemy	Database interaction layer
Image Storage	Cloudinary	Store and serve face photos
Email	SMTP (Gmail / SendGrid)	System notification emails
Report Generation	ReportLab + openpyxl	PDF and Excel exports
Deployment	Railway / Render	Cloud hosting (free tier available)

6.2 Database Schema Overview

Table	Key Fields
institutions	id, name, logo_url, admin_email, created_at
departments	id, institution_id, name, code, faculty
programmes	id, department_id, name, code, duration_years
courses	id, programme_id, lecturer_id, title, code, level, semester, threshold_pct
users	id, email, password_hash, role (admin/lecturer/student), is_active

Table	Key Fields
lecturers	id, user_id, staff_id, department_id, name
students	id, user_id, student_id, programme_id, level, face_encoding
student_courses	id, student_id, course_id (junction table)
sessions	id, course_id, lecturer_id, label, started_at, ended_at, qr_token, qr_expires_at, session_code, is_active
attendance_records	id, session_id, student_id, checked_in_at, method (face/qr), status (present/absent), is_manual_override
notifications	id, user_id, type, message, is_read, created_at

7. Non-Functional Requirements

Category	Requirement
Security	All passwords hashed with bcrypt. All API routes protected with JWT. HTTPS enforced. Session codes expire. QR tokens are single-use per student.
Performance	Attendance check-in must complete within 3 seconds. Dashboard must load within 2 seconds. Face recognition response must be under 5 seconds.
Availability	System must target 99% uptime during active academic semesters.
Scalability	System must support up to 5,000 concurrent users without degradation.
Usability	All key student actions must be completable in under 4 taps/clicks. UI must be fully responsive on mobile browsers.
Privacy	Face encodings (not raw images) stored in the database. Raw face photos deleted after encoding. Compliant with GDPR principles.
Auditability	All manual attendance overrides logged with user, timestamp, and reason. All admin actions logged in the audit trail.

8. Suggested Development Timeline

Phase	Tasks	Duration
Phase 1: Foundation	Project setup, authentication system, role-based routing, database schema design, admin dashboard skeleton.	3 weeks
Phase 2: Admin Module	Department, programme, course, lecturer, and student management. Bulk CSV import. Invitation email system.	3 weeks
Phase 3: QR Attendance	Session creation, QR generation, session code entry, QR scanning, attendance recording.	2 weeks
Phase 4: Face Recognition	Face photo upload, encoding storage, live face detection, server-side matching, fallback to QR.	3 weeks
Phase 5: Dashboards & Reports	Student and lecturer dashboards, attendance charts, PDF/Excel exports, threshold alerts.	2 weeks
Phase 6: Notifications	Email notifications, in-app alerts, weekly summaries, low-attendance warnings.	1 week
Phase 7: Testing & Deployment	Unit testing, integration testing, bug fixes, performance tuning, cloud deployment.	2 weeks

9. Glossary

Term	Definition
Session	A single class meeting for which attendance is being recorded.
Session Verification Code	A 6-character code announced by the lecturer during a session that the student must enter to confirm physical presence before marking attendance.
QR Code	A machine-readable 2D barcode generated per session, scanned by students to record attendance.
Face Encoding	A numerical vector representation of a student's facial features, used for identity matching.
Attendance Threshold	The minimum percentage of sessions a student must attend to remain in good standing (default 75%).
JWT	JSON Web Token — a secure token used to authenticate and authorise user sessions.
Programme	An academic course of study (e.g. BSc Computer Science).
Level	The year of study within a programme (e.g. Level 100, Level 200).
Defaulter	A student whose attendance percentage has fallen below the institution-defined threshold.
Audit Trail	A system log that records all manual changes and admin actions for accountability.

End of Document — Smart Attendance System PRD v1.0